

Manas Agrawal

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EDUCATION

York University

Bachelor of Science in Computer Science

Toronto, Canada

Expected May, 2027

- **Related Coursework:** Databases, Data Structures, Object Oriented Programming, Software Design.

TECHNICAL SKILLS

Languages: Java, Python, C, HTML, CSS, JavaScript, TypeScript, Dart, SQL

Libraries & Frameworks: Spring Boot, .NET, Pandas, NumPy, scikit-learn, React, PyTorch

Tools: Amazon Web Services, Azure, Docker, Kubernetes, Git, Node.js, PostgreSQL, MongoDB, Figma, Power BI

EXPERIENCE

Research Assistant

York University

Toronto, Canada

May 2025 – Present

- Designed **ML models** to predict **wireless signal blockages**, achieving **82% accuracy** and **0.81 F1-score**.
- Implemented **deep learning models** including **CNNs**, **GRUs**, and **Vision Transformers**.
- Built a comprehensive end-to-end pipeline handling **data collection**, **preprocessing**, **feature engineering**, **model training**, and **evaluation** for **multimodal datasets**, enabling accurate blockage prediction.
- Compared multiple architectures and optimized **model inference**, improving prediction speeds by **5 times**.
- Integrated **visual and beam data** to enhance prediction of line-of-sight **blockages in mmWave networks**.

Exam Invigilator

York University

Toronto, Canada

Oct 2024 – Present

- Created a **document automation tool** to streamline document prep for **50+ staff**, reducing manual workload.
- Developed **automated document generation, formatting, and distribution** with a **user-friendly interface** using **Python** and **Excel**, reducing prep time by **90%**.
- Invigilated **200+ exams per term** for students with Accessibility Services, ensuring accommodations.

PROJECTS

StockSprout | *Java, Spring Boot, React, CSS, PostgreSQL*

- Developed a **full-stack paper trading platform** with **real markets**, **virtual funds**, and **portfolios**.
- Built secure **REST APIs** with **Spring Boot**, **JWT**, and **PostgreSQL** for order and user management.
- Implemented **portfolio tracking** and **analytics**, including **gain/loss calculations** and **trend monitoring**.
- Designed an **interactive React frontend** with **dynamic stock charts** and **visualizations** for easy analysis.

Jobo3 | *Python, AngularJS, AWS Lambda, API Gateway, DynamoDB, EventBridge*

- Built a **full-stack job application tracker** to manage applications, statuses, and opportunities efficiently.
- Developed a responsive **AngularJS frontend** with dynamic forms and real-time UI updates.
- Implemented scalable **serverless backend services** using **AWS Lambda**, **API Gateway**, and **DynamoDB**.
- Designed **RESTful APIs** and event-driven workflows with **EventBridge** for secure, reliable data processing.

RayTracer | *Java, Object-Oriented Programming*

- Built a **Ray Tracer** in **Java** using core **OOP principles**, modeling key rendering components.
- Implemented **ray-sphere intersection**, **BSDF material**, and basic **shadow computation** to render 3D scene.
- Designed a **modular structure** with classes like **Camera**, **Ray**, **Sphere**, and **Material** for clean extensibility.

LEADERSHIP EXPERIENCE

Director of Finance

Enactus York

Toronto, Canada

Jan 2026 - Present

- Helped increase funding for the club by over **35%** by collaborations with various clubs.
- Developed **automated budget tracking tools** using **Python** and **Excel** to streamline reporting.
- Tracked **expenses** and maintained **transparent financial reporting** across multiple teams.
- Collaborated with cross-functional teams to optimize **resource allocation** and support project growth.